Supplementary Material S2: Modelling and interpretation protocol

This supplementary material documents the final BPMN and DMN modelling decisions for the selected NVL COPD sections. The tables use a consistent structure across all subsections. Where German wording from the source protocol was present, it was translated into English. Original guideline passages are presented as concise translated excerpts.

Supplement S2a: Modelling and interpretation protocol for NVL COPD Section 3.2

The BPMN modelling decisions for the smoking cessation process are presented in Table S2a-1. The table shows how selected source passages from NVL COPD Section 3.2 were translated into BPMN elements and modelling assumptions. The corresponding DMN structure is documented in Table S2a-2. It links the relevant variables, rule conditions, outputs, and BPMN references used to represent the decision logic of NVL COPD Section 3.2.

Table S2a-1. BPMN modelling decisions for NVL COPD Section 3.2.

Original quote	Translation into English	Modelling issue	Modelling decision and rationale
Abbildung 3: “ Rauchende Patient*innen mit COPD “	Figure 3: 'Smoking patients with COPD.'	The algorithm names the target population but does not define an explicit trigger event.	The target population was modelled as the start event: 'Patient with COPD is currently smoking'. This treats the population statement as the trigger for the smoking cessation process.
3.2.1, Seite 31: “Darüber hinaus wurde auch ein ausführlicher und detaillierter Fragebogen speziell für rauchende Patient*innen mit COPD (siehe Abbildung 2) entwickelt, welcher für eine umfassende Dokumentation herangezogen werden kann.”	Section 3.2.1: A detailed questionnaire for smoking patients with COPD can be used for comprehensive documentation.	The guideline does not clearly specify who documents which information from the smoking habits questionnaire.	The role allocation was treated as a modelling assumption. The patient completes the smoking questionnaire, while the physician records or completes the documentation of smoking habits.
S. 32: „Meist werden die Patient*innen zunächst allgemein auf das Rauchen angesprochen, Risiken werden dargestellt und eine Empfehlung zum vollständigen Rauchstopp ausgesprochen.“	Section 3.2.2: Patients are usually asked about smoking, informed about risks, and advised to stop smoking completely.	Figure 3 refers to motivational counselling, while the text describes brief counselling in more detail. It had to be decided whether these should be separate tasks.	The actions were modelled as one user task, 'Conduct motivational counselling'. The task includes asking about smoking, risk counselling, and recommending complete smoking cessation.
S. 32: „Motivierende Gesprächsführung ist eine zusätzliche Option bei Raucher*innen [23], deren Motivation, mit dem Rauchen aufzuhören, gering ist.“	Section 3.2.2: Motivational interviewing is an additional option for smokers whose motivation to stop smoking is low.	The guideline does not define a formal threshold for 'low motivation'. No score, cut-off, or standard question is provided.	The classification of low motivation remains a clinical judgement. Motivational interviewing was modelled as an optional additional

			step rather than as a mandatory task.
S. 33: „Eine Therapie zur Tabakentwöhnung soll sowohl zur Tabakentwöhnung motivierten rauchenden Patientinnen mit COPD, als auch Patientinnen ohne Entwöhnungswunsch angeboten werden.“	Section 3.2.3: Smoking cessation therapy should be offered both to motivated smoking patients with COPD and to patients without a wish to stop smoking.	Consent is shown in the algorithm, but no formal consent documentation is specified. The loop-back logic had to be represented in BPMN.	The task 'Offer smoking cessation therapy' was placed after counselling for all smoking patients with COPD. After an unsuccessful attempt, the same task is reused as a renewed therapy offer.
Abbildung 3, S. 31: „Einverständnis der Patient*innen“	Figure 3: 'Patient consent.'	The guideline mentions consent but does not specify a formal consent document or a later recall process when consent is missing.	Consent was modelled with an exclusive gateway, 'Consent to therapy?'. If consent is given, the patient proceeds to therapy. If consent is not given, the process loops back to documentation and later counselling.
Abbildung 3, S. 31: „Planung und Einleitung einer (erneuten) kombinierten Tabakentwöhnungsmaßnahme (verhaltenstherapeutisch und medikamentös)“ S. 33 3.2.3: Bei entwöhnungsbereiten Patient*innen mit COPD soll eine kombinierte Therapie ⇑⇑ mit Verhaltenstherapie und medikamentöser Entzugssyndrombehandlungnachdrücklich empfohlen und angeboten werden.	Figure 3 and Section 3.2.3: Planning and initiation of a combined smoking cessation intervention; combined behavioural and pharmacological treatment is strongly recommended.	The guideline does not specify the detailed internal process steps of behavioural therapy or pharmacological withdrawal treatment in this section.	The intervention was modelled as a collapsed sub-process, 'Start combined cessation therapy'. The detailed content is documented as combined behavioural and pharmacological smoking cessation, but not expanded in the main BPMN model.
S. 33: „Als primärer Endpunkt wurde der prozentuale Anteil der Teilnehmer*innen erfasst, welcher dauerhaft oder über einen längeren Zeitraum (≥ 6 Monate) abstinent war.“ S. 33 “Wurde eine verhaltenstherapeutische Intervention mit einer üblichen medizinischen Versorgung (UsualCare) ver- glichen, so zeigte sich eine prolongierte Abstinenz nach 6 Monaten zu Gunsten der Verhaltenstherapie”	Section 3.2.3: The evidence describes prolonged abstinence over 6 months as an outcome.	The 6-month period refers to an evidence endpoint, not necessarily to a mandatory process timer in the algorithm.	The timer was modelled more neutrally as 'After follow-up period' instead of a fixed 6-month BPMN timer.
Abbildung 3, S. 31: „War die Entwöhnungsmaßnahme erfolgreich?“	Figure 3: 'Was the cessation intervention successful?'	The algorithm asks whether the intervention was successful, but does not operationalise the measurement method.	The task was labelled 'Assess cessation success'. This is more neutral than 'Verify abstinence success' because the guideline does not specify a concrete verification method.

Abbildung 3, S. 31: „Dokumentation“ und „Keine weiteren Entwöhnungsmaßnahmen“	Figure 3: 'Documentation' and 'No further cessation measures'.	The algorithm shows an end state after successful cessation and a loop after unsuccessful cessation, but gives no detailed repetition or stopping rules.	The successful path ends with documentation and the end event 'Smoking cessation successful / no further cessation measures required'. The unsuccessful path loops back to counselling and renewed therapy offer.

Table S2a-2. DMN elements and rules for NVL COPD Section 3.2.

DMN element	Variable / condition	Output	Evidence from NVL COPD	Original quote	BPMN
Input	motivation_level = 'low' or 'high'	-	Motivational interviewing is an option for smokers with low motivation.	Motivierende Gesprächsführung ist eine zusätzliche Option bei Raucher*innen, deren Motivation, mit dem Rauchen aufzuhören, gering ist. [...] Die Rauchenden entwickeln im gemeinsamen Gespräch durch wertfreie Abwägung der Vor- und Nachteile eine zunehmende Änderungsmotivation. (Kap. 3.2.2)	The va BPMN 'Motiva the rel rule lo
Input	consent_to_therapy = true/false	-	Smoking cessation therapy should be offered, but therapy requires patient consent.	Eine Therapie zur Tabakentwöhnung soll Patient*innen ohne Entwöhnungswunsch angeboten werden. Das Angebot darf nicht zur Folge haben, nicht aufhörbereite Patient*innen zu stigmatisieren. (Empfehlung 3-3, Kap. 3.2.3)	The va corres BPMN 'Conse
Input	intensive_cbt_feasible = true/false	-	Intensive behavioural interventions may not be available everywhere; lower-intensity evidence-based offers should then be used.	Intensive verhaltenstherapeutisch-orientierte Interventionen sind im deutschen Versorgungskontext nicht flächendeckend verfügbar. In diesen Fällen sollen direkt andere evidenzbasierte Beratungs- und Unterstützungsangebote angeboten werden. (Kap. 3.2.3, Empfehlung 3-5)	The va routing intens data o task c check
Input	prior_cessation_attempt = true/false	-	The algorithm allows renewed planning and initiation of combined smoking cessation after a failed attempt.	War die Entwöhnungsmaßnahme erfolgreich? [...] Planung und Einleitung einer (erneuten) kombinierten Tabakentwöhnungsmaßnahme (verhaltenstherapeutisch und medikamentös). (Algorithmus Abb. 3, Empfehlung 3-2)	The va true w re-ente unsuc
Output	intervention	Recommended cessation intervention	Combined behavioural and pharmacological treatment is the preferred intervention for motivated patients with COPD.	Kombination aus verhaltenstherapeutischen Maßnahmen und pharmakologischen Verfahren: verlängerte Abstinenz nach 6–12 Monaten (RR 2,53; 4 RCTs, n=1429, Evidenzqualität hoch). (Empfehlung 3-4)	The ou the su path a rule ta
Output	recommendation_grade	Recommendation strength	The combined therapy has a strong recommendation; behavioural single or group	Empfehlungsgrad ⇑⇑ (starke Empfehlung) für kombinierte Therapie; ⇑ für verhaltenstherapeutische Einzel-/Gruppenintervention. Basiert auf Cochrane-Review (van Eerd et al.) und	This is output docum purpos

			interventions are also recommended.	systematischer Update-Recherche. (Empfehlung 3-4, 3-5)	
Rule 1	consent_to_therapy = false	No therapy initiated	Therapy cannot proceed without consent.	Einverständnis der Patient*innen [nein] → keine weiteren Maßnahmen. Die Empfehlung und Empfehlungsstärke beziehen sich ausschließlich auf das Angebot der Therapie. (Algorithmus Abb. 3, Empfehlung 3-3)	The n conse back t and p couns
Rule 2	motivation_level = 'low' AND consent_to_therapy = true AND prior_cessation_attempt = false	Motivational interviewing + offer therapy	Low motivation triggers motivational interviewing as a supportive first-contact option.	Motivierende Gesprächsführung: Ziel ist der Aufbau intrinsischer Motivation zur Verhaltensänderung. [...] Die Rauchenden werden nicht von außen zum Rauchstopp motiviert oder gedrängt. (Kap. 3.2.2)	The B routes motiva intervi re-offe
Rule 3	motivation_level = 'low' AND consent_to_therapy = true AND prior_cessation_attempt = true	Motivational interviewing + renew therapy offer	After a failed cessation attempt, motivational support and a renewed offer are modelled.	Planung und Einleitung einer erneuten kombinierten Tabakentwöhnungsmaßnahme [...] nach gescheitertem Versuch. (Algorithmus Abb. 3)	The re 'Cessa = No' prior_ to true
Rule 4	motivation_level = 'high' AND consent_to_therapy = true AND intensive_cbt_feasible = true AND prior_cessation_attempt = false	Combined therapy	For cessation-ready patients, combined behavioural and pharmacological therapy is strongly recommended.	Bei entwöhnungsbereiten Patient*innen mit COPD soll eine kombinierte Therapie mit Verhaltenstherapie und medikamentöser Entzugssyndrombehandlungnachdrücklich empfohlen werden. (Empfehlung 3-4, ⇑⇑)	This m collap 'Start cessa
Rule 5	motivation_level = 'high' AND consent_to_therapy = true AND intensive_cbt_feasible = true AND prior_cessation_attempt = true	Repeat combined therapy	The algorithm allows renewed combined treatment after an unsuccessful attempt.	Planung und Einleitung einer erneuten kombinierten Tabakentwöhnungsmaßnahme (verhaltenstherapeutisch und medikamentös). (Empfehlung 3-2, Algorithmus Abb. 3)	This re BPMN with prior_ set to
Rule 6	motivation_level = 'high' AND consent_to_therapy = true AND intensive_cbt_feasible = false	Brief counselling / telephone counselling / online programme	If intensive behavioural treatment is not feasible, lower-intensity evidence-based support should be offered.	Falls Patient*innen intensive verhaltenstherapeutische Maßnahmen nicht wahrnehmen können, stehen folgende weniger intensive Angebote zur Verfügung: Kurzberatung, Telefonberatung (BZgA 0800 8313131), Online-Programme, Selbsthilfe-Materialien. (Kap. 3.2.3, Empfehlung 3-5, ⇑)	A sepa branch intens suppo recom case.

Supplement S2b: Modelling and interpretation protocol for NVL COPD Section 3.3

The main process question was whether an e-cigarette should be used to support tobacco abstinence in a smoking patient with COPD.

The BPMN model uses two pools: Patient with COPD and Healthcare provider. The healthcare provider pool is divided into the lanes Physician, Behavioral counselling, and Follow-up/monitoring. This structure was chosen because the guideline describes e-cigarette use only as a strictly conditional exception. Patients may consider using e-cigarettes, but the clinical side must provide information about potential risks, assess whether the required conditions are met and ensure continuous monitoring. Behavioral counselling is represented as an additional supportive component.

For the BPMN/DMN modelling, Section 3.3 was divided into five logical blocks: the lack of COPD-specific evidence of benefit, the potential for harm, the guideline position against using e-cigarettes as the primary cessation attempt, the exception criteria for supportive use, and the need for additional counselling and monitoring. This structure translates the narrative recommendation into a restrictive decision and process logic. BPMN represents the care process, while DMN checks whether the conditions for the exception decision are fulfilled.

The five logical blocks used to structure the e-cigarette modelling are summarised in Table S2b-1. The table explains how the narrative recommendation was divided into evidence, risk, exclusion, exception, and counselling or monitoring components. The BPMN modelling decisions for the e-cigarette process are presented in Table S2b-2. The table documents how triggers, prerequisite checks, counselling, and continuous monitoring were represented in the process model. The DMN eligibility rules for e-cigarette-supported tobacco abstinence are presented in Table S2b-3. The table is intended to be understandable independently of the BPMN model and therefore includes the primary cessation attempt as a separate exclusion rule. The BPMN modelling decisions for the exacerbation-focused medication algorithm are shown in Table S2c-1. The table describes how the algorithm was transformed into start events, gateways, user tasks, business rule tasks, and end states. The DMN decision for adherence and inhalation technique before medication escalation is documented in Table S2c-2. The table defines the inputs and outputs that determine whether escalation may proceed or whether training or counselling is required first. The DMN decision for ICS indication before escalation to triple therapy is documented in Table S2c-3. The table specifies how eosinophil counts and an asthma component were used to determine whether ICS escalation is guideline-concordant. The DMN decision for ICS review and potential de-escalation is documented in Table S2c-4. The table captures the priority rule for pneumonia under ICS, the eosinophil-based de-escalation logic, and the asthma-protection rule.

Table S2b-1. Logical blocks used for BPMN/DMN modelling of NVL COPD Section 3.3.

Logical block	Source content	BPMN / DMN relevance
1. Evidence base: no COPD-specific benefit evidence	The guideline reports that no publications were found that investigated the effects of e-cigarettes specifically in patients with COPD.	This supports a restrictive process model. E-cigarettes are not represented as a standard therapy.
2. Risk and harm profile	The guideline describes potential adverse effects, variable product composition, potentially toxic emissions, cardiovascular effects, and additional risk from dual use.	This supports the BPMN task 'Counsel patient about known risks' and the monitoring tasks for adverse effects and health risks.
3. Guideline position: not a primary cessation attempt	Because of insufficient evidence and uncertain long-term effects, the guideline group does not consider e-cigarettes an option for a primary cessation attempt.	This supports the gateway 'Primary cessation attempt?' and the exclusion path ending with 'E-cigarette not initiated as primary cessation attempt'.
4. Exception pathway: only under prerequisites	If e-cigarettes are considered to support tobacco abstinence, the guideline requires documented failure or refusal of other evidence-based measures, risk counselling, simultaneous cessation of tobacco use, and continuous medical monitoring and support.	This supports the prerequisite assessment sub-process and the DMN decision 'Determine e-cigarette eligibility'.
5. Behavioural counselling and nicotine-content information	The guideline states that e-cigarette therapy should ideally include behavioural counselling and notes that nicotine content must be labelled.	Behavioural counselling was modelled as an additional supportive component. The nicotine-content note can be used as an annotation for counselling or monitoring, but not as a separate decision point.

Table S2b-2. BPMN modelling decisions for NVL COPD Section 3.3.

Original quote	Translation into English	Modelling issue	Modelling decision

„Gleichwohl ist der Leitliniengruppe bewusst, dass manche Raucher*innen dieses Produkt nutzen, um ihren Tabakkonsum zu beenden.“	The guideline acknowledges that some smokers use e-cigarettes to stop tobacco use.	It is unclear whether the process is triggered by the patient or by the physician.	The process starts in the patient pool with 'Patient considers e-cigarette use for smoking cessation' and a message event communicating interest. The patient pool is kept simple because the section does not describe further patient actions.
„Falls der Einsatz der E-Zigarette zur Unterstützung der Tabakabstinenz erwogen wird [...]“	If the use of e-cigarettes to support tobacco abstinence is considered...	The guideline does not define an explicit trigger for the start of the clinical assessment.	A message flow from the patient pool to the healthcare provider pool starts the physician process with 'Request for counselling received'. This makes the counselling trigger visible.
„Aufgrund der aktuell unzureichenden Datenlage [...] sieht die Leitliniengruppe in der E-Zigarette keine Option für einen primären Entwöhnungsversuch.“	The guideline does not consider e-cigarettes an option for a primary cessation attempt.	The distinction between a primary cessation attempt and exceptional supportive use had to be formalised.	A physician task 'Clarify intended role of e-cigarette' is followed by the exclusive gateway 'Primary cessation attempt?'. The gateway is exclusive because only one path can apply.
„[...] sieht die Leitliniengruppe in der E-Zigarette keine Option für einen primären Entwöhnungsversuch.“	The guideline does not consider e-cigarettes an option for a primary cessation attempt.	It is unclear whether an alternative cessation process should be modelled after rejecting e-cigarette use.	The yes-path ends with 'Recommend against primary e-cigarette cessation attempt' and the end event 'E-cigarette not initiated as primary cessation attempt'. Alternative cessation measures are outside Section 3.3.
„Falls der Einsatz der E-Zigarette zur Unterstützung der Tabakabstinenz erwogen wird, dann nur unter folgenden Voraussetzungen:“	If e-cigarettes are considered, then only under the following prerequisites...	The prerequisites are listed, but not defined as an explicit process sequence.	The no-path from the primary attempt gateway leads to an expanded sub-process 'Assess prerequisites for e-cigarette-supported abstinence'. The sub-process groups the prerequisites without implying that the guideline defines this exact sequence.
„Bei dokumentiertem Versagen oder Ablehnung anderer evidenzbasierter Maßnahmen,“	Documented failure or refusal of other evidence-based measures.	The guideline does not specify how or where documented failure or refusal is technically verified.	A user task 'Check documented failure or refusal of evidence-based measures' was included. The condition is also checked in DMN as a Boolean input.
„nach Aufklärung über bekannte Risiken,“	After counselling about known risks.	Risk counselling is mandatory, but its exact content and scope are not fully operationalised.	A user task 'Counsel patient about known risks' was included. The task is mandatory, but the detailed content remains based on the risk description in the guideline.
„bei gleichzeitiger Beendigung des Tabakkonsums und“	With simultaneous termination of tobacco use.	Dual use is described as a risk, but not as a separate algorithmic decision node.	A user task 'Confirm simultaneous termination of tobacco use' was included to exclude parallel use of tobacco cigarettes and e-cigarettes.
„bei kontinuierlichem ärztlichem Monitoring und Begleitung.“	With continuous medical monitoring and support.	Monitoring is required, but interval, duration, stopping rules, and outcome criteria are not specified.	A task 'Plan continuous medical monitoring and support' was included, followed by a monitoring sub-process. Time intervals and stopping criteria remain documented modelling assumptions.
„Falls der Einsatz [...] erwogen wird, dann nur unter folgenden Voraussetzungen: [...]“	Only under the following prerequisites...	The four prerequisites form an AND logic, but representing all of them as BPMN gateways would make	The eligibility decision was moved to DMN. BPMN represents the process flow, while DMN represents the strict

		the process difficult to read.	AND logic of the mandatory prerequisites.
„Falls der Einsatz der E-Zigarette zur Unterstützung der Tabakabstinenz erwogen wird, dann nur unter folgenden Voraussetzungen:“	Only under the following prerequisites...	The negative decision path must be explicit because e-cigarette use is allowed to be considered only when all prerequisites are met.	If the DMN decision is negative, BPMN routes to 'Do not initiate e-cigarette-supported abstinence' and the end event 'E-cigarette not initiated'.
„Falls der Einsatz der E-Zigarette zur Unterstützung der Tabakabstinenz erwogen wird, dann nur unter folgenden Voraussetzungen:“	Only under the following prerequisites...	After a positive DMN decision, BPMN needs a visible transition from prerequisite assessment to monitored use.	If all prerequisites are fulfilled, the task 'Initiate monitored e-cigarette abstinence' is used. 'Monitored' indicates that use is linked to medical monitoring and support.
„Idealerweise bedarf die Therapie mittels E-Zigarette einer verhaltenstherapeutischen Beratung.“	E-cigarette therapy should ideally include behavioural counselling.	The wording 'ideally' is weaker than the mandatory prerequisites, so it was unclear whether counselling should be a required condition.	Behavioural counselling was modelled as a collapsed sub-process and annotated as recommended, not as a mandatory eligibility criterion.
„bei kontinuierlichem ärztlichem Monitoring und Begleitung.“	Continuous medical monitoring and support.	It is unclear when monitoring ends and which outcomes should trigger continuation or discontinuation.	A looping monitoring sub-process was modelled with scheduled follow-up, review of tobacco abstinence and e-cigarette use, adverse-effect assessment, and medical support. Follow-up intervals and outcome criteria remain assumptions.

Note: Although the primary cessation attempt is already excluded in the BPMN model, it was also included in the DMN table as the first exclusion rule. This was done to keep the eligibility logic complete and understandable as a standalone decision table. The variable primary_cessation_attempt therefore acts as an additional safeguard in the DMN model, even though the same question is already represented earlier as a BPMN gateway.

Table S2b-3. DMN eligibility rules for e-cigarette-supported tobacco abstinence in NVL COPD Section 3.3.

Original quote	Translation into English	Variable / condition	Output	Reason	Modelling decision
„Aufgrund der aktuell unzureichenden Datenlage, und auch wegen der noch nicht abschätzbaren gesundheitlichen Langzeitfolgen für Patient*innen mit COPD, sieht die Leitliniengruppe in der E-Zigarette keine Option für einen primären Entwöhnungsversuch.“	E-cigarettes are not an option for a primary cessation attempt.	primary_cessation_attempt = true	Do not initiate	E-cigarette is intended as primary cessation attempt.	This exclusion was modelled as the first rule. Although this question is already represented as a BPMN gateway, it is also included in DMN to keep the eligibility table complete as a standalone decision table.
„Bei dokumentiertem Versagen oder Ablehnung anderer	Documented failure or refusal of	primary_cessation_attempt = false AND	Do not initiate	No documented failure or	'Documented' is required, but the guideline

evidenzbasierter Maßnahmen,“	other evidence-based measures.	documented_failure_or_refusal = false		refusal of other evidence-based measures.	does not specify the data source or form of proof. The condition was modelled as a Boolean input.
„nach Aufklärung über bekannte Risiken,“	After counselling about known risks.	primary_cessation_attempt = false AND documented_failure_or_refusal = true AND risk_counseling_completed = false	Do not initiate	Risk counselling has not been completed.	Risk counselling is mandatory but not fully operationalised. DMN checks only whether it has been completed; the counselling itself remains a BPMN task.
„bei gleichzeitiger Beendigung des Tabakkonsums und“	With simultaneous termination of tobacco use.	primary_cessation_attempt = false AND documented_failure_or_refusal = true AND risk_counseling_completed = true AND simultaneous_tobacco_cessation = false	Do not initiate	Simultaneous termination of tobacco use is not confirmed.	The Boolean input prevents parallel use of tobacco cigarettes and e-cigarettes.
„bei kontinuierlichem ärztlichem Monitoring und Begleitung.“	With continuous medical monitoring and support.	primary_cessation_attempt = false AND documented_failure_or_refusal = true AND risk_counseling_completed = true AND simultaneous_tobacco_cessation = true AND continuous_monitoring_planned = false	Do not initiate	Continuous medical monitoring and support is not planned.	DMN checks whether monitoring is planned; the monitoring process itself is represented in BPMN.
„Falls der Einsatz der E-Zigarette zur Unterstützung der Tabakabstinenz erwogen wird, dann nur unter folgenden Voraussetzungen: Bei dokumentiertem Versagen oder Ablehnung anderer evidenzbasierter Maßnahmen, nach Aufklärung über bekannte Risiken, bei gleichzeitiger Beendigung des Tabakkonsums und bei kontinuierlichem ärztlichem Monitoring und Begleitung.“	Only under the following prerequisites: documented failure/refusal, risk counselling, simultaneous tobacco cessation, and continuous medical monitoring and support.	primary_cessation_attempt = false AND documented_failure_or_refusal = true AND risk_counseling_completed = true AND simultaneous_tobacco_cessation = true AND continuous_monitoring_planned = true	May consider under strict conditions	All mandatory prerequisites are fulfilled.	Because the guideline states 'only under the following prerequisites', the conditions were modelled as strict AND logic.

Supplement S2c: Modelling and interpretation protocol for NVL COPD Section 5.1 (Exacerbations that have occurred)

This section documents the modelling of the medication algorithm for patients in whom exacerbations that have occurred are predominant.

Table S2c-1. BPMN modelling decisions for NVL COPD Section 5.1.

Original quote	Translation into English	Modelling issue	Modelling decision
Seite 62: In Abbildung 4 steht als Einstieg in diesen Pfad: „Aufgetretene Exazerbationen vorrangig“. Seite 63: „Die rechte Seite stellt mögliche Therapien bei Patient*innen da, welche vermehrt Exazerbationen in der Anamnese oder im Krankheitsverlauf erlebt haben …“	Figure 4 starts the right-hand pathway with 'Exacerbations that have occurred are predominant'. The narrative describes therapies for patients with increased exacerbations in history or disease course.	The algorithm does not define a true start event, but assumes an already established clinical classification.	The BPMN model starts with 'Exacerbations are predominant'. It does not model the prior assessment of exacerbations, but starts from the already established clinical state.
Seite 62: „Therapienaiv? … ja … nein“. Fußnote 1 ergänzt: „Bei bereits mit der Kombination LABA/ICS vorbehandelten Patient*innen ist die Überprüfung der Indikation für eine ICS-Gabe zu prüfen.“	Figure 4: 'Therapy-naive? yes/no'.	The yes/no split is explicit, but 'therapy-naive' is not formally operationalised in the excerpt.	An exclusive gateway 'Therapy-naive?' was modelled. For a later DMN implementation, the condition would need clearer operational criteria, such as no relevant prior long-term inhaled therapy.
Seite 62-63: • Algorithmus bei „ja“: „LAMA oder LAMA + LABA“ • Narrativ: „In den unterschiedlichenTherapiestufen sind teils mehrereAlternativen pro Kasten … aufgeführt. Welche davon im individuellen Fall in Frage kommen, müssen Ärztinnen und Patientinnen vor dem Hintergrund der persönlichen Umstände sowie der zuerwartenden Wirkungen und Nebenwirkungen entscheiden.“	For therapy-naive patients, the algorithm lists 'LAMA or LAMA + LABA'. The narrative states that several alternatives may be shown and that physicians and patients choose based on individual circumstances and expected effects and side effects.	The algorithm shows two alternatives in one box, but they are not parallel actions.	The alternatives were modelled as one physician task, 'Select initial therapy: LAMA or LAMA + LABA'. This reflects a clinical choice rather than simultaneous treatment paths.
Seite 62: Der Algorithmus unterscheidet nachTherapienaiv? den nein-Pfad und zeigtdarunter die Eskalationsstufen mit„Vorbehandelt mit:“ und „Eskalationauf:“. Dabei werden als VorbehandlungLAMA, LAMA + LABA oder LABA + ICS berücksichtigt. Außerdem gilt: „Vor jeder Eskalation ist die Adhärenz zurmedikamentösen Therapie sowie die Inhalationstechnik zu prüfen.“ Seite 65: „Die vorliegende Evidenz stützt die Kombination von LAMA und LABA alsnächste Eskalationsstufe.“	The algorithm distinguishes therapy-naive from pretreated patients and then shows escalation levels.	After 'Therapy-naive? = yes', the excerpt does not show direct escalation in the same pathway.	The yes-path ends with 'Initial exacerbation-focused therapy selected'. This does not exclude later escalation, but places it outside the initial pathway or in a later clinical contact.
Seite 62:	For 'Therapy-naive? = no', the algorithm	The algorithm and narrative	The task was labelled 'Select / continue LAMA +

Algorithmus bei „nein“: „LAMA + LABA“ Seite 65: Narrativ zur LAMA/LABA-Eskalation: „Die vorliegende Evidenz stützt die Kombination von LAMA und LABA alsnächste Eskalationsstufe.“	leads to 'LAMA + LABA'. The narrative supports LAMA/LABA as the next escalation step.	use LAMA + LABA as a target or intermediate step for pretreated patients.	LABA or LAMA'. It is not modelled as a final end state because subsequent escalation logic may still apply.
Seite 62: „Vor jeder Eskalation ist die Adhärenzzur medikamentösen Therapie sowiedie Inhalationstechnik zu prüfen.“	Before each escalation, adherence to medication and inhalation technique must be checked.	It had to be decided whether this check should be repeated before each escalation step or placed once before the escalation pathway.	The check was modelled centrally before the pretreatment gateway. This keeps the BPMN readable while preserving the requirement that all escalation paths depend on this prerequisite.
Seite 62: „Vorbehandelt mit:“ mit den Optionen„LAMA“ sowie „LAMA + LABA oder LABA + ICS“	The algorithm lists pretreatment with LAMA, LAMA + LABA, or LABA + ICS.	The algorithm is closer to a table mapping pretreatment to escalation than to a classical process split.	This was translated into the gateway 'Current pretreatment?' with three outgoing paths: LAMA, LAMA + LABA, and LABA + ICS.
Seite 62: „Vorbehandelt mit: LAMA“ →„Eskalation auf: LAMA + LABA“.	Pretreated with LAMA -> escalation to LAMA + LABA.	This escalation is explicitly visible in the algorithm.	The step was modelled as the user task 'Escalate from LAMA to LAMA + LABA' rather than being hidden entirely in DMN.
Seite 62: Algorithmus-Fußnote 1: „Bei bereits mit der Kombination LABA/ICS vorbehandelten Patient*innen ist die Überprüfung der Indikation für eine ICS-Gabe zu prüfen.“ Seite 65: Narrativ: „… empfiehlt … die Indikation für eine ICS-Gabe regelmäßig zu überprüfen und dies abzusetzen, wenn die Indikation nicht (mehr) besteht.“	For patients pretreated with LABA/ICS, the indication for ICS must be checked. The narrative states that the indication for ICS should be reviewed regularly and ICS discontinued if the indication no longer exists.	The algorithm mentions triple therapy, but the ICS indication is specified mainly through footnotes and narrative text.	The check was modelled as the business rule task 'Check ICS indication before triple therapy'. Thresholds such as eosinophil counts are better represented in DMN than in BPMN.
Seite 62: „Vorbehandelt mit: LAMA + LABA oder LABA + ICS“ → „Eskalation auf: LAMA + LABA + ICS“.	Pretreated with LAMA + LABA or LABA + ICS -> escalation to LAMA + LABA + ICS.	Triple therapy must not appear as an automatic consequence of LAMA + LABA; it depends on the prior ICS check.	The task 'Escalate from LAMA + LABA to LAMA + LABA + ICS' follows only after a positive or not excluded ICS indication.
Seite 62: Fußnote 1: „Bei bereits mit der Kombination LABA/ICS vorbehandelten Patient*innen ist die Überprüfung der Indikation für eine ICS-Gabe zu prüfen.“ Seite 67–68: Empfehlung 5-2 fordert einen Absetzversuch von ICS bei	For patients pretreated with LABA/ICS, the indication for ICS must be checked.	This check applies specifically to patients already receiving LABA/ICS and should not be placed before all paths.	The task 'Review ICS indication' was modelled only in the LABA + ICS path.

Eosinophile< 100 Zellen/µl und fehlenden asthmatischen Komponenten oder bei Pneumonie unter ICS			
Seite 62: „Vorbehandelt mit: … LABA + ICS“ → „Eskalation auf: LAMA + LABA + ICS“.	Pretreated with LABA + ICS -> escalation to LAMA + LABA + ICS.	The algorithm groups LAMA + LABA and LABA + ICS as pretreatments for the same escalation, but LABA + ICS requires an additional ICS review.	The LABA + ICS path was separated in BPMN so that the specific ICS review remains visible.
Seite 62: Algorithmus-Fußnote 3: „Die Gabe von Roflumilast ist bei Patient*innen mit COPD möglich, die trotz Therapie wiederholt exazerbieren, dem ‚chronische Bronchitis‘-Phänotyp zuzuordnen sind und eine FEV1 < 50% haben.“ Seite 69-70: Narrativ: „… Roflumilast als letzte Eskalationsstufe zu einer Triple-Therapie … wenn wegen erhöhter Exazerbationsgefahr weiterhin Handlungsbedarf besteht. In einigen Fällen ist Roflumilast auch statt ICS eine Option als Add-on zu einer LAMA/LABA-Kombination, nämlich, wenn ICS-Kontraindikationen bestehen …“	Roflumilast may be given as an add-on in patients who continue to exacerbate despite therapy, have the chronic bronchitis phenotype, and have FEV1 < 50%.	The algorithm says 'if applicable, Roflumilast', so it is not a mandatory escalation step.	Roflumilast was modelled as the optional gateway 'Roflumilast add-on eligible?' with clear eligibility criteria.
Seite 62: Algorithmus: „Ggf. Roflumilast (als Add On zu LAMA + LABA oder LAMA + LABA + ICS)“. Seite 69-70: Narrativ: „… Roflumilast als letzteEskalationsstufe zu einer Triple-Therapie …“ und „… auch statt ICS eineOption als Add-on zu einer LAMA/LABA-Kombination … wenn ICS-Kontraindikationen bestehen …“	Roflumilast may be added to LAMA + LABA or LAMA + LABA + ICS.	Roflumilast can connect to two different treatment targets.	A shared task 'Add Roflumilast to LAMA + LABA or LAMA + LABA + ICS' was placed after the paths converge instead of creating two separate add-on tasks.
Seite 62: Roflumilast ist als Add-on zu LAMA + LABA oder LAMA + LABA + ICS vorgesehen.	Roflumilast is shown as an add-on option.	Adding Roflumilast changes the medication plan beyond selecting inhaled therapy.	This path ends with 'Exacerbation-focused medication plan adjusted'.
Seite 62: Der Algorithmus endet je nach Pfad in LAMA + LABA, LAMA + LABA + ICS oder optional Ggf. Roflumilast.	The algorithm ends depending on the path in LAMA + LABA, LAMA + LABA + ICS, or optional Roflumilast.	Without Roflumilast, the treatment plan remains the selected or escalated inhaled therapy.	This path ends with 'Exacerbation-focused medication plan selected' rather than 'adjusted'.

Table S2c-2. DMN decision table for adherence and inhalation technique before escalation.

DMN element	Variable / condition	Output	Evidence from NVL COPD	BPMN reference
Input	goodAdherence = true/false	-	Treatment adherence must be checked before escalation.	The input is evaluated through the BPMN task 'Check adherence and inhalation technique'.
Input	correctInhalationTechnique = true/false	-	Incorrect use of inhalation systems must be excluded before dose or step escalation.	The input is evaluated through the same BPMN task.
Output	escalationAllowed	true/false	This controls whether medication adjustment can continue.	The output feeds the process gateway after the check.
Output	adherenceAction	Required action	This gives the concrete action when training or counselling is required.	The output can route to a follow-up user task.
Rule_Adh_1	goodAdherence = true AND correctInhalationTechnique = true	escalationAllowed = true; adherenceAction = 'Proceed to pretreatment evaluation'	Adherence and technique are adequate.	Medication pretreatment can be evaluated.
Rule_Adh_2	correctInhalationTechnique = false	escalationAllowed = false; adherenceAction = 'Train inhalation technique and re-evaluate later'	Inhalation errors take priority over medication change.	BPMN should route to training and later re-evaluation.
Rule_Adh_3	goodAdherence = false AND correctInhalationTechnique = true	escalationAllowed = false; adherenceAction = 'Counsel patient on adherence and re-evaluate later'	Adherence counselling is required before escalation.	BPMN should route to counselling and later re-evaluation.

Table S2c-3. DMN decision table for ICS indication before triple therapy.

DMN element	Variable / condition	Output	Evidence from NVL COPD	BPMN reference
Input	eosinophils	-	Blood eosinophil counts guide the indication for ICS use in patients with exacerbations.	The value is evaluated in the business rule task 'Check ICS indication before triple therapy'.
Input	asthmaComponent = true/false	-	An asthma component supports ICS indication.	This may be provided by a previous diagnostic task or gateway.
Output	icsIndicated	true/false	The output indicates whether adding ICS is guideline-concordant.	It feeds the BPMN gateway 'ICS indication present?'.
Rule_Ics_1	eosinophils >= 300	icsIndicated = true	At eosinophil counts of 300 cells/microlitre or higher, clinical benefit of ICS is supported.	BPMN can proceed towards triple therapy.

Rule_Ics_2	eosinophils < 300 AND asthmaComponent = true	icsIndicated = true	An asthma component supports ICS even with lower eosinophil counts.	BPMN can proceed towards triple therapy.
Rule_Ics_3	eosinophils < 300 AND asthmaComponent = false	icsIndicated = false	If neither asthma nor elevated eosinophils are present, ICS should not be added.	BPMN should not proceed to ICS escalation.

Table S2c-4. DMN decision table for ICS review and de-escalation.

DMN element	Variable / condition	Output	Evidence from NVL COPD	BPMN reference
Input	eosinophils	-	Eosinophil count guides de-escalation decisions for existing ICS therapy.	Evaluated in 'Review ICS indication'.
Input	pneumoniaUnderIcs = true/false	-	Pneumonia under ICS requires critical re-evaluation.	Can be assessed in the clinical risk pathway.
Input	asthmaComponent = true/false	-	An asthma component protects against inappropriate ICS withdrawal.	Provided by prior diagnostic assessment.
Output	maintainIcs	true/false	The output controls whether ICS should be maintained or discontinued.	It routes to de-escalation or continuation of therapy.
Rule_Rev_1	pneumoniaUnderIcs = true	maintainIcs = false	This is a priority rule: if pneumonia occurs under ICS, ICS should be discontinued.	Hit policy FIRST was used to prioritise this safety rule.
Rule_Rev_2	eosinophils < 100 AND pneumoniaUnderIcs = false AND asthmaComponent = false	maintainIcs = false	For eosinophils below 100 cells/microlitre without asthma, the guideline recommends discontinuing ICS.	BPMN routes to the ICS de-escalation path.
Rule_Rev_3	eosinophils 100-299 AND pneumoniaUnderIcs = false AND asthmaComponent = false	maintainIcs = true	The NVL does not define an explicit threshold-based rule for this intermediate range. This creates a formalisation boundary because DMN requires a clear output.	The rule documents the modelling assumption that clinical judgement predominates in this range.
Rule_Rev_4	eosinophils >= 100 AND pneumoniaUnderIcs = false	maintainIcs = true	At intermediate or higher eosinophil counts, benefit is assumed to outweigh withdrawal unless other risks apply.	BPMN routes to continuation of ICS.
Rule_Rev_5	pneumoniaUnderIcs = false AND asthmaComponent = true	maintainIcs = true	If an asthma phenotype is present and there is no acute pneumonia, ICS should not be discontinued.	This protects against inappropriate de-escalation.